Supplementary Information

for "Sample-and-hold governance: review intervals as a design variable in resource stability"

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Scope note. This file records methods, proofs, inputs, per-stock results, reconstruction details, sensitivity outputs, and unresolved evidence. "Reproduced" means that a deposited computation or table was rerun and matched within the stated tolerance; "archived" means an existing output is transcribed or preserved; "reconstructed" means a new implementation was specified and run; and "prospective" means the design has not been executed. A deterministic grid result is not a continuous-parameter proof. A cross-sectional proxy is not a measured institutional treatment.

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S1 Claim, evidence, and limitations register

Table S1: Evidence-status register for Paper07 v54.

Claim / object	Status	Auditable record	Limitation retained in v53
Forward invariance	Proved under stated assumptions	Proposition S2.1	Positivity and effort bounds do not imply persistence, viability, or stability.
Rapid-review consistency	Proved on fixed finite horizons under regularity	Lemma S2.2	Does not establish stability for any fixed positive T_{r}.
Logistic-core crossing	Reproduced deterministic computation; full nonlinear-flow Jacobian checked independently	Core output, baseline tables, nonlinear-flow check in S11	Illustrative, uncalibrated parameter vector; no invariant-circle continuation or nonlinear bifurcation theorem.
Joint six-parameter boxes	Re-run for all 128 corners on 20,001-point grid	Full-grid JSON, script, and logs in bundle	Non-crossing is dense-grid numerical stability, not a proof between grid points; boxes are not confidence sets.
Stage-structured response window	Archived trajectory classifications plus new reconstruction	Stage log; reconstruction and comparison in S9.2	Archived solver configuration/histories absent; new map does not reproduce all archived windows.
42-stock SSB screen	Re-executed, multiplicity controlled	RAM v4.66 inputs and screen outputs in source bundle	Screening diagnostic, not a governance-sign or causal test; AR(1) null is incomplete for forcing and regime changes.
T_{r}-ranked test	Script and outputs recovered and rerun	Ranked-test script, per-stock CSV, reproduction log	Only two group-coded proxy values; all below windows; no approach gradient identifiable. Reproduced opposite-direction two-group association is confounded and non-causal.
ADH / fishing-mortality verification	Re-executed locally on recovered RAM cohorts	ADH/F logs and recovered cohort inputs	Matching archived tables verifies arithmetic and inputs, not the validity of ADH as a management response measure.
White-noise power table	17 cells assembled from archived records	Power CSV and JSON	Fifteen cells come from archived run log; two cod $H = 400$ cells from follow-up JSON. The complete driver did not finish.
Nonlinear comparator trajectories	Archived deterministic and 10-seed summaries	Archived comparator CSVs	The single-run script is retained; the 10-seed summary CSV omits its own noise/seed metadata and is not independently rerun here.
32-system case search	Curated inventory / documentary review	Case table and query log in S4	Not a systematic review; no qualifying case found is not evidence of non-existence.
Northern cod partition	Descriptive assessment-table reconstruction	DFO table values and cod figure	No causal allocation among governance, ecology, predation, climate, recruitment, and assessment. Constrained-M values are not reproduced.
Distributive constraints	Measurement-level inventory only	S6	No operational floor or licence-holder panel; no normative threshold inferred.
Cross-sector designs	Prospective	S7 and S14.3	Not executed; no effect estimates.

Change in the interpretation of the T_{r} test. The recovered test is executable and its output has been regenerated. It reports both (a) no identifiable window-approach gradient because the only assigned values are

0.5 and 1.0 years while the sampled windows are 4 and 8 years, and (b) reproducible two-group associations in the opposite direction from a simple approach prediction. Those statements are not contradictory. The first is an identification limit; the second describes a proxy-group contrast. Neither supports an institutional causal interpretation. The Peru/Chile clustering makes regional ecology, including ENSO-related quasi-periodicity, a plausible confound, but the present test does not establish that ENSO caused the contrast.

Reproducibility. The executable ranked-test script, per-stock output table, and reproduction log are included in the source bundle. The local rerun reproduces the logged values to the displayed precision. The inference reported here follows the current executable record and its two-proxy limitation.

S2 Mathematical statements, proofs, and the review-map operator

S2.1 Forward invariance

Let the flow on a review interval be

$$\dot{N} = rN(1 - N/K) - qE(t)N, \qquad \dot{Z} = \{\Phi(qE(t)N - S(N)) - Z\}/\tau_m,$$

where $r, K, q, \tau_m > 0$, $\Phi \geq 0$, and $E(t) \in [0, E_{\max}]$. At each review, the effort command is projected onto $[0, E_{\max}]$. Between reviews effort is either held at an admissible endpoint or interpolated convexly between consecutive admissible commands.

Proposition S2.1 (Forward invariance of the sampled state space)**.** *If $N(0) \geq 0$, $Z(0) \geq 0$, and $E_0 \in [0, E_{\max}]$, then every solution that exists remains in $N \geq 0$, $Z \geq 0$, and $E \in [0, E_{\max}]$.*

Proof. The effort statement follows from projection and convexity. For a fixed admissible effort path, the stock equation is locally Lipschitz and has the factorised form $\dot{N} = N\{r(1 - N/K) - qE(t)\}$. If $N(t_0) > 0$, then

$$N(t) = N(t_0) \exp\left(\int_{t_0}^{t} [r(1 - N(s)/K) - qE(s)]\, ds \right) > 0;$$

if $N(t_0) = 0$, uniqueness gives the zero solution. For the signal, variation of constants gives

$$Z(t) = e^{-(t-t_0)/\tau_m} Z(t_0) + \frac{1}{\tau_m} \int_{t_0}^{t} e^{-(t-s)/\tau_m} \Phi(qE(s)N(s) - S(N(s)))\, ds \geq 0.$$

These facts hold interval by interval. Projection places each next command in the same effort interval, completing the induction. The logistic field also gives $\dot{N} \leq 0$ at $N = K$, so on a finite interval $N(t) \leq \max\{N(t_0), K\}$. No lower bound strictly above zero, persistence, viability, or stability follows from this proposition. □

S2.2 Finite-horizon rapid-review consistency

Lemma S2.2 (Finite-horizon rapid-review consistency)**.** *Let $\dot{u} = G(u)$, $u = (N, Z, E)$, be the no-queue continuous system, with the effort component evolving under the same local controller law. Suppose G is continuously differentiable with bounded derivative on a compact neighbourhood of the trajectories over a fixed horizon $[0, T]$; assessments are contemporaneous and exact; projection is inactive on that neighbourhood; and no deployment queue, stochastic jump, or unmodelled reset is added. Let Ψ_{T_r} be either the exponential frozen-assessment or forward-Euler flow-then-update map. Then, uniformly on the compact set,*

$$\Psi_{T_r}(u) = u + T_r G(u) + O(T_r^2),$$

and, for a common initial state, the sampled trajectory differs from the continuous trajectory by at most $C_T T_r$ at review times with $nT_r \leq T$. This is a fixed finite-horizon statement; it does not imply that any fixed positive T_r has the same stability classification as the continuous system.

Proof. Taylor expansion of the held ecological flow and endpoint effort update gives the displayed one-step expansion. The exact flow of G has the same first-order expansion, so the local defect is $O(T_r^2)$. If L bounds the Lipschitz constant of G, the global error $\epsilon_n = u_n - u(nT_r)$ satisfies

$$\|\epsilon_{n+1}\| \leq (1 + LT_r)\|\epsilon_n\| + CT_r^2.$$

Discrete Gronwall gives $\max_{nT_r \leq T} \|\epsilon_n\| \leq C_T T_r$, establishing the claimed uniform convergence at review times. No infinite-horizon convergence or finite-interval stability result follows. □

Let $\widehat{z}_{n+1} = \widehat{Z}_{n+1} - Z^*$ be the endpoint assessment deviation after the ecological state has flowed for one review interval with effort held at E_n. The nonlinear exponential update integrates $d\mathcal{E}/ds = F_B(\mathcal{E}, \widehat{Z}_{n+1})$ for $0 \le s \le T_\mathrm{r}$, holding that assessment fixed. Linearizing this controller flow at the interior equilibrium, with $C_E = \partial_E F_B(E^*, Z^*)$ and $C_Z = \partial_Z F_B(E^*, Z^*)$, gives

$$e^{\exp}_{n+1} = e^{C_E T_\mathrm{r}} e_n + C_Z \frac{e^{C_E T_\mathrm{r}} - 1}{C_E} \widehat{z}_{n+1},$$

with limiting coefficient $C_Z T_\mathrm{r}$ when $C_E = 0$. The corresponding forward-Euler update is $e^{\mathrm{Euler}}_{n+1} = e_n + T_\mathrm{r}(C_E e_n + C_Z \widehat{z}_{n+1})$. Their difference is second order as $T_\mathrm{r} \to 0$, but can be material at finite T_r. In the monodromy, this post-flow command update is composed with the held ecological flow, which supplies the dependence of $\widehat{z}_{n+1}$ on the pre-review state.

S2.3 Local stability and crossing conventions

Let $X^* = (N^*, Z^*, E^*)$ be an interior fixed point of the pre-review map $X^-_{n+1} = \mathcal{P}_{T_\mathrm{r}}(X^-_n)$. The local multipliers are the eigenvalues of $D\mathcal{P}_{T_\mathrm{r}}(X^*)$. Local asymptotic stability requires all multipliers to have modulus below one. A crossing is counted when a computed multiplier reaches the unit circle along the scanned interval branch. A complex-pair crossing, a real $+1$ crossing, and a real -1 crossing are distinguished; the values in this paper are numerical crossings refined by bisection and high-precision evaluation.

The baseline values at $T_\mathrm{r} = 1$ year are $\rho = 1.0003506565$ for exponential frozen-assessment mobilising, 1.0005452356 for forward-Euler mobilising, 0.9838095183 for exponential frozen-assessment protective, and 0.9837962340 for forward-Euler protective. The exponential-update mobilising crossing is a complex pair with eigenvalue approximately $0.98464047 \pm 0.17459425i$, modulus one, and angle 0.175494 radians. Forward-Euler mobilising crosses as a complex pair at 47.535982 years and at a real multiplier -1 at 79.142688 years. Forward-Euler protective crosses at a real multiplier -1 at 2.306354 years. The exponential protective update has no crossing on the tested $[0.2, 200]$ interval and has maximum sampled radius 0.996740. The forward-Euler crossings are properties of that distinct command update, not numerical estimates of the exponential-update boundary.

S2.4 Nonlinear crossing check and unresolved invariant-curve claim

A fresh finite-difference audit now evaluates the full nonlinear map: DOP853 integrates (N, Z) under effort held over the review interval, then integrates the nonlinear effort-rate ODE with the endpoint assessment fixed for duration T_r. Projection is inactive at the interior fixed point. The fixed-point residual is zero at printed precision. At relative finite-difference step $h = 3 \times 10^{-5}$, the Jacobian agrees with the exponential monodromy to maximum absolute error 6.93×10^{-11} at $T_\mathrm{r} = 1$ year and 1.67×10^{-9} at the monodromy crossing; the corresponding spectral radii are 1.000350656546 and 1.000000000031. The finite-difference crossing estimates across $h = 10^{-4}, 3 \times 10^{-5}, 10^{-5}, 3 \times 10^{-6}, 10^{-6}$ are shown below. These are numerical step-size diagnostics, not uncertainty intervals.

Table S2: Finite-difference crossing of the full nonlinear plant plus exponential frozen-assessment controller flow. The reference linear-monodromy crossing is 6.5012949481 years.

Relative step h	Finite-difference crossing (yr)
10^{-4}	6.501299288
3×10^{-5}	6.501295007
10^{-5}	6.501293726
3×10^{-6}	6.501289941
10^{-6}	6.501276055

The earlier archived normal-form script differentiated a map with a nonlinear plant but a linearized exponential controller update; its "supercritical" classification is not a normal-form result for the full nonlinear controller flow and is not treated as established here. No invariant-circle continuation or independent orbit confirmation was completed. The local crossing is therefore verified, but a nonlinear bifurcation theorem or stable invariant-circle claim is not made. The procedure, full outputs, and code are retained in `work/nonlinear_flow_check_v53.py` and `work/nonlinear_flow_check_v53.json`.

S3 Data provenance, coding rules, and reproducibility register

S3.1 Input and record inventory

Table S3: Primary analysis inputs and status. Repository-relative paths are given in the v53 source bundle.

Object	Source / version	Local file family	Use and limitation
Forage-fish SSB	RAM Legacy Stock Assessment Database v4.66	`src/p7_empirical/ram_target_stocks.csv` and screen outputs	42-stock primary screen; the extracted series and derived summary are preserved.
ADH cohort	RAM v4.66 and archived v4.44 cohorts	`src/p7_empirical/recovered/ram_adh_fisheries/`	Local recomputation of buffer-years and stock/group summary; version differences retained.
Fishing mortality	RAM fisheries-ADH input plus cohort records	`src/p7_empirical/recovered/ram_adh_fisheries/`	F placement and quartile verification; one stock does not match the last-F record.
Northern cod	DFO CSAS SAR 2016/026 assessment table	`src/p7_empirical/recovered/ices_cod_27_5a.csv`	SSB and reported M descriptive plot; not a raw mortality-decomposition archive.
Stage-structured plant	Archived parameter and trajectory records	`src/p7_empirical/recovered/`; `src/results/`	Archived windows and new reconstruction have distinct status.
Logistic sensitivity	Archived one-at-a-time battery	`src/p7_empirical/recovered/sens_table.csv` and log	Wider $\pm 5\%/\pm 10\%$ record; distinct from the v53 $\pm 2\%$ rerun.
Case search	Curated primary-source query record	`src/p7_empirical/u5_case_table.csv`, `src/p7_empirical/u5_query_log.md`	32 rows; not an exhaustive review.
Power grid	Archived run log plus two-cell follow-up	`work/power_table_white_noise.csv/.json`	17-cell table assembled; complete simulation driver run did not finish.

S3.2 Reproduction entry points

The source bundle README provides command lines and expected outputs. Principal entry points are:

- `work/p07_core.py --grid 4001`: logistic fixed point, local multipliers, crossing scans, the v53 $\pm 2\%$ one-at-a-time sensitivity, and the baseline record; use `--joint-only --grid 20001` for the full 128-corner rerun;
- `src/p7_empirical/ram_crosssection.py` and `src/p7_empirical/tr_ranked_test.py`: stock-level spectra and the T_{r}-ranked test; run from `src/p7_empirical/` so the relative RAM input path resolves;
- `work/screen_battery_v53.py`: screen sensitivity battery; run from `src/p7_empirical/`;
- `work/adh_verify_v53.py` and `work/qcal_fplacement_v53.py`: local ADH, F-placement, and calibration checks using the recovered cohort inputs;
- `work/nonlinear_flow_check_v53.py`: full nonlinear-plant/controller-flow finite-difference check against the local monodromy;
- `work/make_figures_v53.py`: the joint sensitivity figure from the retained full-grid JSON;
- `work/make_rho_figure_v53.py`: the baseline multiplier plot with explicit exponential-update labels;
- `work/make_supplement_tables.py`: per-stock, case, power, comparator, and F-placement LaTeX tables.

Exact working directories, dependency versions, seeds, and stdout/log files are retained in the source bundle rather than inferred from this list.

S3.3 Local ADH and fishing-mortality verification

Recovered cohort inputs resolved a previous missing-file issue. The ADH script was rerun locally against the recovered files. Recomputed ADH values agree with deposited values to numerical precision, and archived descriptives and cohort/group summaries reproduce. A fresh 20,000-draw seed-11 resampling run placed 2.33% of broader-pool median draws below the archived cohort median of 1.7902 buffer-years; the archived run gives 2.12%. This checks the resampling calculation and shows a similar tail fraction, not a calibrated probability that any fishery is unsustainable.

The F-calibration and placement script reproduces the 42-stock screen cohort. For 43 intersected stocks, fisheries-ADH F agrees with last-F for 42; `SPRBLKGSA29` is the exception. For the broader pool of 454 stocks, the F quartiles and all 13 taxonomic-group quartiles match the archived summaries. Among 38 screen-cohort stocks with F series, the mean of stock-level F medians is 0.3663 (IQR 0.2701–0.7243); broader-pool F_{now} has median 0.2143 (IQR 0.0800–0.4639). These are descriptive calibration and placement checks, not proof that F or an ADH buffer is a valid proxy for institutional pressure.

S4 Structured case-search record

S4.1 Eligibility criteria and adjudication rule

The inventory contains 32 candidate systems. It is a curated query-driven search, not a systematic review. A candidate was eligible only if all four criteria were supported by evidence:

1. an operative, repeatedly responsive governance rule (not a one-time cap, ban, fixed path, coordination deadlock, or nominal rule with no effective response);
2. an independently dateable response or implementation lag at the scale relevant to the ecological outcome;
3. no competing ecological or structural driver supported for that same outcome, or comparative evidence that institutional feedback predicts the outcome better than prespecified competing mechanisms; and
4. individual-resource or station-level observations adequate to order the response signal, decision, implementation, and ecological effect.

The third condition is not satisfied merely because an institutional mechanism is plausible. If ecology, recruitment, climate, predation, cohort resonance, or assessment structure is supported for the same outcome and no discriminating comparison exists, the case is coded as confounded or unresolved. "Yes" in the table means that the available record supports the item; "no" means the criterion fails; a dot denotes no determination or inapplicability; "partial" denotes incomplete evidence. The binding-fail column is the recorded failure, not a composite score.

Twenty cases are coded as having a responsive rule, with one additional case coded partial; 21 rows record criterion (iii) as unmet. Two produced-capital examples are out of the regenerating-resource scope. No row satisfies all four criteria simultaneously. The closest candidates—Bangkok groundwater and La Mancha Oriental groundwater—remain confounded by subsidence/rainfall and crop/water substitution, respectively. These decisions should be read as limits of this inventory, not as proof that no qualifying system exists.

Table S4: Curated 32-system inventory and four-criterion dispositions. A dot denotes no assessment or not applicable; the binding-fail column records the criterion or scope exclusion that prevents full eligibility.

ID	System	Domain	C1	C2 lag	C3	C4	Binding fail	Evidence note
1	Baltic sprat (SPRAT22-32 EU MAP)	fisheries	yes	yes (˜1-1.5 yr)	no	yes	iii	Annual TAC responsive; variance is low-frequency trend/regime (no robust cycle); tau-window screen row; RAM SSB series
2	European anchovy Adriatic (GFCM MAP)	fisheries	no	.	.	.	i	2022-2026 fixed predetermined reduction path: no deficit feedback (GFCM/44/2021/20; GFCM/48/2025/2)
3	European anchovy Bay of Biscay	fisheries	yes	yes (˜1-1.5 yr)	no	yes	iii	Annual TAC; recruitment/environment-driven variability; closed 2005-2009
4	Peruvian anchoveta	fisheries	yes	yes (sub-annual)	no	yes	iii	ENSO driver: 3.7-yr peak Granger-caused by ENSO (section 3.7 battery)
5	Pacific sardine (US /Mexico)	fisheries	yes	yes (˜0.5-1 yr)	no	yes	iii	Annual HCR; environmentally forced regime variability
6	Atlantic menhaden (ASMFC)	fisheries	yes	yes (˜1-1.5 yr)	no	yes	iii	Annual quota; environmentally forced recruitment variability
7	Gulf menhaden (GSMFC)	fisheries	yes	yes (˜1-1.5 yr)	no	yes	iii	Annual quota (5-state commission); environmentally forced variability
8	North Sea herring (ICES)	fisheries	yes	yes (˜1-1.5 yr)	no	yes	iii	Annual TAC; recruitment/climate confounds
9	Atlanto-Scandian herring	fisheries	no	.	.	.	i	Coastal-state sharing deadlock since ˜2014; unilateral quotas above advice: coordination failure not a rule
10	NE Atlantic mackerel / blue whiting	fisheries	no	.	.	.	i	Coastal-state deadlock (MSC suspended 2019-2020): no operative responsive rule
11	Iberian sardine (PT/ES)	fisheries	yes	yes (˜1-1.5 yr)	no	yes	iii	Annual TAC + rebuilding measures; environmentally forced variability
12	South Africa anchovy /sardine (OMP)	fisheries	yes	yes (˜1-1.5 yr TAC)	no	yes	iii	Annual TAC within OMP (rule review 4-5 yr is not effort-response lag); environmentally forced
13	Namibian sardine/anchovy	fisheries	yes	yes (˜1-1.5 yr)	no	yes	iii	Annual TAC; environmentally forced variability
14	Chilean anchovy / jack mackerel	fisheries	yes	yes (˜1 yr)	no	yes	iii	Annual TAC + seasonal management; environmentally forced variability
15	Japanese sardine	fisheries	yes	yes (˜1 yr)	no	yes	iii	Annual TAC; regime-shift-driven variability
16	West African sardinella	fisheries	no	no	.	.	i,ii	Annual TACs exist but enforcement/implementation weak and irregular (CECAF): no dateable responsive lag
17	NZ QMS pelagics	fisheries	no	.	.	.	i	TACs held between sparse 3-10 yr assessments (57% never changed): discontinuous not delayed-continuous (Mace et al. 2013/2014)
18	Barents / Icelandic capelin	fisheries	yes	yes (¡1-1 yr)	no	yes	iii	Annual quotas + in-season surveys; dynamics cod-driven/environmentally forced
19	North Sea sandeel	fisheries	yes	yes (˜1-1.5 yr)	no	yes	iii	Annual TAC seasonal; environmentally forced + seabird-driven closures
20	India/Philippines /Indonesia small pelagics	fisheries	no	.	.	.	i	No TAC; seasonal closures/mesh regulations only: fails responsive-feedback criterion
21	Icelandic cod (1995 HCR)	fisheries	yes	yes (0.2-0.3 yr impl. lag)	no	yes	iii	25% HCR responsive; cohort resonance alternative (section 3.7/S4); post-rule SSB CV 0.387 (audit-confirmed)
22	Icelandic haddock (2013 HCR)	fisheries	yes	yes (˜1 yr TAC cycle)	no	yes	iii	35%-of-B45cm+ HCR adopted Apr 2013 (Iceland govt); recruitment-forced variability (R CV 0.89)
23	Sheridan-6 groundwater	groundwater	no	.	.	yes	i	55 acre-inch block allocation over five-year first period: not continuously adjusted feedback (S4)
24	Bangkok groundwater	groundwater	yes	yes (post-1999 response)	no	yes	iii	Pumping declines after 1999 but subsidence/recovery confounded by heterogeneous consolidation + rainfall (S4)
25	La Mancha Oriental groundwater	groundwater	.	.	no	.	iii	2019-2023 extraction relapse; attribution complicated by crops/ET/surface-water substitution (S4)
26	Murray-Darling surface water	surface water	yes	yes (annual announcements)	no	yes	iii	Allocations responsive to storage/inflows (SA DEW; BoM NWA 2024) but availability is rainfall-driven: climate confound

ID	System	Domain	C1	C2 lag	C3	C4	Binding fail	Evidence note
27	US BLM rangeland grazing	rangeland	partial	no	.	.	ii	Responsive adjustments only via recent flexibility pilots (IM 2018-109; outcome-based grazing); historically rigid permits with NEPA-gated irregular lags: no dateable response lag
28	Scandinavian moose harvest	wildlife	yes	yes (annual/3-5 yr plans)	no	yes	iii	Responsive quotas (SE annual county; NO 3-5 yr plans) but recruitment climate-driven (hot dry springs/snow) + wolf predation confounds
29	Norwegian salmon aquaculture (TLS)	aquaculture	yes	yes (dated evaluations 2017+)	no	yes	iii	Biomass responsive to lice-induced wild-smolt mortality (TLS 13 areas) but lice signal biologically/environmentally forced + farm economics joint: institutional channel not isolated
30	Spruce budworm forestry	forestry	no	.	.	.	i	30-40 yr cycle is endogenous predator-prey outbreak dynamics (Ludwig Jones Holling 1978): no institutional feedback at all
31	Hog/livestock cycles	produced capital	.	.	.	.	scope	Unconfounded delay oscillators but no autonomously regenerating stock: outside the resource-model scope (section 3.7/S4)
32	Electricity-capacity cycles	produced capital	.	.	.	.	scope	Unconfounded delay oscillators but no autonomously regenerating stock: outside the resource-model scope (section 3.7/S4)

S4.2 Search provenance and query-log reconciliation

The retained `u5_query_log.md` is dated 2026-09-11. It records (i) repository tree and code searches; (ii) targeted extraction of relevant release files; (iii) RAM Legacy v4.66 re-execution and the cod/haddock data audit; (iv) ICES stock-data pulls for northern cod and haddock; and (v) primary-source web searches for water allocation, grazing, moose harvest, salmon-lice governance, and Icelandic harvest-control rules. The table's first 20 rows were re-adjudicated under the four criteria rather than mechanically inherited from an earlier delay-window screen. Row-level notes and search queries are preserved verbatim in the source bundle.

The query log's search R4 found a memo that called the T_{r}-ranked test "never executed." That was a historical snapshot of the analysis status, not the final repository state: the executable test script and outputs were subsequently recovered and re-run. The discrepancy is documented in S1 and the source reconciliation note; it does not change the case-screening result.

S5 Ecosystem context, proxy risks, and case-level audits

S5.1 Peru and Chile: a plausible confound is not an identified cause

The seven stocks assigned $T_{\mathrm{r}} = 0.5$ years in the ranked test are Peruvian or Chilean anchovy stocks with annual quotas and group-described in-season responsive closures. The code does not verify each stock's specific closure trigger, decision date, or state-assessment cadence. The shorter proxy is therefore a regional group classification. It is not a measured stock-specific treatment.

A retained anchoveta catch series and prior analysis notes associate a roughly 3.7-year feature with ENSO-related environmental variation. That association motivates a confound check, but the v53 ranked test does not re-estimate a causal ENSO model on a fully verified primary-source governance panel. The per-stock AR(1) surrogate reproduces fitted red-noise persistence but not quasi-periodic forcing. Consequently, the observed negative T_{r}-rank association and the higher two-group null-excess in Peru/Chile are compatible with regional forcing, cohort composition, assessment differences, or other omitted factors; they do not establish which factor produced the contrast. The correct result remains: no identifiable approach gradient, plus an opposite-direction proxy-group association with unresolved confounding.

This matters because ENSO and longer regime shifts can structure anchoveta abundance, and oceanographic change has been documented off Peru (Chávez et al., 2003; Gutiérrez et al., 2009). Those references provide ecological context, not an instrumental variable or proof that ENSO caused the v53 contrast.

S5.2 Icelandic cod and haddock: numerical audit of an earlier forensics note

A recovered cod/haddock audit used current-vintage ICES standard-graph source-data tables, covering cod 1955–2026 and haddock 1979–2026. For 1996–2023, cod SSB has $n = 28$, CV 0.394, catch CV 0.150, fishing-mortality CV 0.199, and recruitment CV 0.208. The previously archived cod CV of 0.387 is confirmed as the same post-1995 concept under an earlier vintage; the current-vintage value is 0.394. For haddock over the same years, SSB CV is 0.359 and recruitment CV is 0.894. The earlier value 0.143 for haddock is not reproduced under standard SSB-CV definitions: detrended-residual CV is 0.345, log-difference volatility is 0.171, and the observed SSB CV is 0.359. That number is therefore not used in v53.

A post-1995 detrended cod spectrum has its strongest peak near 19.58 years; the 10–15-year band contains 14.9% of the power under the retained calculation. The peak is not a stable, narrow institutional signature and the series is trend-dominated. The numerical audit confirms the earlier coding note's broad environmental/recruitment confound, but neither this CV comparison nor the spectral summary identifies an HCR effect. The haddock recruitment variability result is descriptive and does not adjudicate the governor's review cycle.

S5.3 Northern cod ecosystem evidence

The northern-cod SSB and mortality values used in Fig. 3 and Table S20 are copied from the DFO CSAS 2016/026 assessment record. The 1991–1995 crash window has assessment SSB values 735, 382, 101, 31, and 10 kt for 1991–1995, respectively; the full transcribed table is in Supplementary S13.3. The post-1995 portion contains later declines and recoveries, so it is called a non-recovery *record*, not a monotone trend. Tam and Bundy (2019) provide mass-balance descriptions of harp-seal and capelin changes on the Newfoundland and Labrador Shelf; those are ecosystem context only. No seal-predation mortality rate is estimated here, and the constrained-M decomposition is not reproduced.

S5.4 Model-class diagnostic for scalar autonomous surplus production

A simple scalar autonomous surplus-production equation with constant removals has the form

$$\dot{S} = f(S) - C, \qquad f(S) = \frac{r}{K - \mathfrak{s}}(K - S)(S - \mathfrak{s}),$$

where $0 < \mathfrak{s} < K$, $C \geq 0$, and S is the modeled stock. The phase-line shape of f limits the equilibrium structure. If $C > 0$, the vector field at zero is $-C$, so the state requires an explicit nonnegativity constraint or a different low-stock removal rule. For extra mortality $M_x \geq 0$, write $f_M(S) = f(S) - M_x S = S(h(S) - M_x)$ with $h(S) = f(S)/S$. The zero equilibrium remains; positive equilibria solve $h(S) = M_x$. When $0 < M_x < \max h$, the quadratic h can have two positive roots, including a lower threshold shifted above $\mathfrak{s}$; at the maximum the roots coalesce and above it positive equilibria disappear.

This is a model-class observation, not an empirical test of northern cod. The assessment series is not an error-free trajectory and removals and mortality are not constant, so no contradiction between this scalar model and the real stock is claimed. The lemma says what this class would imply if its assumptions held; it does not establish those assumptions.

S5.5 Interpretive guardrails

The contexts above are used to motivate competing mechanisms and the need for discriminating data. They do not turn the spectral screen, the review-interval proxy, a coefficient of variation, or a visual event sequence into evidence of institutional causation. The classification of an environmental mechanism as "plausible" is intentionally weaker than a causal finding.

S6 Distributive constraints and measurement-level social indicators

S6.1 Candidate constituency and observed measurement

For the northern-cod case, a candidate worst-off constituency is registered inshore harvesters and licence holders. Available community-level data do not measure that population directly. The mismatch is retained rather than resolved by substituting a convenient scalar:

Table S5: Measurement mismatches in the cod social-indicator record.

Object	Candidate definition	Available measure	Mismatch	Data needed
Population	Registered licence holders	Census-subdivision residents	Different population	Licence-holder microdata or administrative panel
Income	Fishing income of the named constituency	All-resident sector income, including aquaculture and processing	Includes non-fishing income and other activities	Administrative fishing-income records with linkage and privacy protection
Floor	Specified instrument and cutoff (I_k, c_k)	No operational cutoff	Normative and measurement rule absent	Named instrument, reference period, deflator, cutoff, and authority

Statistics Canada Tables 38-10-0167-01 and 38-10-0168-01 report community-level series for 43 fishing-dependent census subdivisions; the mean share of employment income from fishing falls from 32.2% to 25.6% between 2016 and 2021. The statistic is all-resident employment income, not an inshore licence-holder panel. It includes aquaculture and processing categories and cannot by itself identify which households rely on the cod fishery or who bears a policy transition.

A measured floor is a pair (I_k, c_k): an instrument and its cutoff. For groups G and times t, define margin $m_k(G,t) = I_k(G,t) - c_k$. A componentwise constraint is violated if any required margin is negative; it does not permit a favourable aggregate indicator to compensate for a household or group falling below a different floor. No such floor is operationalized in v53. Non-decline is a normative choice, not automatically an empirically justified threshold. A defensible instrument must state baseline, cohort, inflation and purchasing-power treatment, uncertainty, attrition, acceptable variation, authority, and structural diversification.

S6.2 External measurement frameworks and limits

The global Multidimensional Poverty Index (Alkire and Foster, 2011) and the World Bank Poverty and Inequality Platform are potential reference frameworks, but their release and vintage must be locked before use. Neither

substitutes for a local distributional record of who pays for persistence, who may use remaining resource diversity, whose knowledge counts, or how future persons are represented. These issues are left open as measurement and governance questions, not filled by a composite score. No distributional causal effect is estimated in this manuscript.

S7 Prospective empirical programme and falsification rules

The designs below are proposals, not completed tests. No registration identifier, locked protocol, or field intervention is claimed. Each should be registered before outcome inspection, with data provenance, estimand, decision rule, and deviations recorded.

S7.1 Design 1: governance-event panel

Unit and data. Resource–jurisdiction–assessment cycle. Link dated raw observation, assessment data freeze, advice release, committee decision, legal instrument, implementation, compliance, and ecological response. Record date uncertainty and source document for each event. Separate T_{obs}, T_{assess}, T_{r}, τ_{dec}, τ_{dep}, and τ_{eco}; do not impute one "governance lag" from another.

Estimand. Distribution of effective responsive-review intervals and the impulse/phase response of command, extraction, and stock to a measured assessment innovation, conditional on operating-model class.

Identification. The panel is descriptive until independent variation or a design-based assumption identifies the effect. Use date ranges where records are incomplete; do not encode a scheduled annual TAC as a 0.5-year response merely because in-season closures exist without trigger evidence.

Falsification. A hypothesised sample-and-hold mechanism is weakened if decisions precede the information they are said to respond to, if implementation does not follow the decision, or if the predicted command–stock phase order fails on held-out cycles.

S7.2 Design 2: quasi-experimental schedule variation

Identify an external schedule change plausibly independent of current ecological innovations. Candidate approaches include event studies, difference-in-differences, synthetic controls, interrupted time series, and state-space interventions, selected according to the institutional change and comparison structure. Define treatment as a change in responsive decision opportunities, not simply a change in implementation lag. A nominal calendar reform is not a treatment if the same rule is still applied to the same information on the same effective schedule.

Estimand. Change in extraction and ecological outcomes attributable to a change in the review operator, with pre-specified direct and lagged effects.

Identification checks. Pre-trends, concurrent rule and data changes, ecological shocks, serial correlation, spillovers, anticipatory behaviour, implementation/compliance, and measurement-vintage changes. Use placebo dates and placebo stocks where feasible.

Falsification. If estimated schedule variation is explained by an ecological shock or if outcomes move before the schedule change, the causal interpretation is rejected or narrowed.

S7.3 Design 3: out-of-sample mechanism comparison

Fit competing models with distinct causal orderings: (i) ecological forcing only; (ii) cohort resonance or recruitment structure; (iii) institutional feedback with a pre-specified rule sign and sample-and-hold operator; (iv) combined ecology and governance; and (v) a non-periodic or regime-shift null. Freeze training and test windows, transformations, priors, and scoring rules before fitting. Compare held-out log predictive density or another proper score; report phase-order checks and posterior predictive discrepancies, not only in-sample spectral fit.

Estimand. Relative out-of-sample predictive performance of the institutional mechanism versus its alternatives, and the estimated timing/gain parameters conditional on the winning model class.

Falsification. The delay mechanism is weakened when it fails to predict held-out phase ordering or performs worse than an ecological/cohort competitor. It is displaced for that case when the competitor predicts held-out observations better under the pre-specified score. A significant spectrum by itself is not the decision rule.

S7.4 Design 4: randomized human-in-the-loop experiment

In a laboratory or advisory-interface setting, randomize review cadence and feedback sign or timing in a common renewable-resource simulator. Participants receive a controlled stream of noisy observations and implement

decisions through the simulated institution. Randomize treatment assignment, mask the hypothesis where possible, and measure state, command, decision latency, revisions, and errors. A live fishery is not randomized by this proposal.

Estimand. Intention-to-treat effects of assigned review cadence and rule architecture on stability, transient stock loss, command volatility, and decision quality.

Falsification. The proposed gain–phase signature is weakened if randomized cadence does not alter outcomes in the predicted direction or if observed behavioural adaptation eliminates the modeled distinction. Any laboratory result remains externally limited until field validation.

S7.5 Design 5: closed-loop management strategy evaluation

Construct an MSE with an operating-model ensemble that crosses review interval, decision lag, deployment lag, observation error, rule gain and sign, stock productivity, age structure, environmental forcing, assessment structure, compliance, and process-noise regime. Compare, at minimum, a responsive extractive rule, a responsive protective rule, and a fixed-plan controller; keep catch/effort constraints and objectives identical where meaningful. Use common random numbers across policies and report both local spectral margins and long-run performance metrics.

Estimands. Probability of breaching biological reference points, expected yield and variability, recovery time, implementation burden, distributional indicators, and regret relative to an explicitly stated benchmark.

Falsification. The result does not generalize beyond the MSE if rankings reverse across plausible operating models or if the claimed stability signature does not predict the prespecified performance outcomes. Robustness is reported across model classes, not just parameter draws from one operating model.

S7.6 Cross-cutting preregistration and displacement discipline

Every design should state whether it tests mechanism, prediction, or policy performance. It should lock data vintage, response windows, rule coding, exclusions, the unit of analysis, multiplicity, uncertainty, and an alternative-model set before outcomes are inspected. The cod case contributes two dated governance events (moratorium announcement 2 July 1992; reopening 26 June 2024 with an 18 kt TAC), but these dates alone do not supply a counterfactual or a governance lead. A fast ecological response does not establish adequacy; annual SSB data do not establish a governance lag.

Three broader hypotheses motivate, but are not tested by, the present work: (a) observation aggregation can change the apparent feedback phase; (b) governance phase ordering can be more diagnostic than periodicity alone; and (c) a substitution certificate can identify when continuous-delay and sampled-data models cease to be interchangeable. Testing them requires the event-linked, high-frequency, independently varied data described above.

Boundary-limited design implications. The governance-clock principles are best read as modelling and evaluation requirements, not policy prescriptions: (i) include the review interval, hold/update law, and fixed-plan versus responsive controller as explicit factors in management-strategy evaluation; (ii) compare controller sign and update operator before transferring a stability threshold; (iii) measure observation, assessment, decision, deployment, and ecological-response intervals separately; and (iv) if emergency triggers or circuit-breakers are proposed for structural breaks, model them as a distinct hybrid-control feature and compare them against a no-trigger baseline. The present results do not establish a universally stabilising cadence, that protective rules generally tolerate institutional rigidity, that extractive rules require a particular review frequency, or that emergency triggers improve outcomes. The annual-review result and finite-grid protective non-crossings are model-specific numerical observations, not general management conclusions.

S8 Model parameters and reconstruction conventions

S8.1 Illustrative logistic hold-map baseline

The equilibrium is $N^* = 89.5518830$, $Z^* = 0.0693147181$, $E^* = 2.08962340$, and $F^* = qE^* = 0.00208962340\,\mathrm{yr}^{-1}$. The exploitation ratio is $qE^*/r = 0.10448117$. The coefficients are not posterior estimates or quantities extracted from a specific stock. In particular, comparisons of model F^* with empirical assessed F are scale-placement diagnostics only; model effort units and stock-specific reference points are not jointly calibrated.

At the interior equilibrium, the local effort-law gains from the independent core are $(C_E, C_Z)_{\text{mobilising}} = (-0.05951801, 1.78501871)$ and $(C_E, C_Z)_{\text{protective}} = (-0.85033614, -1.66170190)$. The exact machine-precision record is in `work/results.json`.

Table S6: Logistic-core and controller parameter vector. Values are model conventions; no field calibration is claimed.

Symbol	Meaning	Baseline	Status / units
r	Intrinsic growth rate	0.020	yr^{-1}; illustrative
K	Logistic carrying capacity	100	Model abundance units; illustrative
q	Catchability	0.001	Model effort-to-mortality scale
η	Mobilising effort-law gain	0.914	Inherited Candidate-A convention; uncalibrated
$E_{\max}$	Upper effort bound	30	Model effort units
δ_0	Effort-law offset gain	0.010	Distinct from softplus shift δ
Z_{ref}	Signal reference scale	1	Signal units
Δ_{ref}	Deficit reference scale	1	Signal units
τ_m	Signal-memory timescale	5	Years
k	Softplus sharpness	10	Signal-map units inverse
δ	Shift in floored softplus	$\log(2)/10 = 0.0693147$	$Z^* = \delta$ at zero deficit
η_p	Protective gain	η	Same numerical gain as mobilising baseline
E_0	Protective target scale	$E^*(Z_{\mathrm{ref}} + \delta)/Z_{\mathrm{ref}}$	Chosen to share the interior rest

S8.2 Stage-structured reconstruction parameterization

Table S7: Frozen ecological values in the preregistered stage reconstruction. The scale $A_0 = 100$ is a convention, not a fitted abundance.

Class	M (yr^{-1})	Recruitment delay τ (yr)	Steepness h	Source / status
Anchovy	0.90	1	0.75	Pauly and Tsukayama (1987); IFOP range; default steepness convention
Sprat	0.40	2	0.75	ICES sprat keyrun / maturity ogive; default steepness convention
Cod	0.20	5	0.75	ICES fixed-M convention; DFO northern-cod maturity; default steepness convention
Slow stock	0.045	25	0.75	Branch (2001) orange-roughy range; default steepness convention

For this reconstruction, $s_A = s_J = e^{-M}$, $A_0 = 100$, and Beverton–Holt recruitment parameters are $c = 4h/(1-h)$, $\beta = (c-1)/A_0$, and $\alpha = c(1-s_A)/s_J$. Primary catchability is $q = 0.001$, with a declared sensitivity at $q = 0.1$; controller parameters are carried over unchanged from the logistic core. A separate steepness sensitivity uses $h \in \{0.6, 0.9\}$. No stage parameter was tuned against archived response windows.

S8.3 Notation and time scales

Table S8: Notation used in the main text and Supplementary Information.

Symbol	Definition
N	Logistic-core resource abundance.
A, J	Adult and juvenile states in the distinct two-stage reconstruction.
Z	Nonnegative filtered institutional deficit signal.
E	Effort, held between reviews and projected to $[0, E_{\max}]$.
$S(N)$	Logistic surplus production $rN(1-N/K)$. Northern-cod SSB is a separate measured quantity and is not substituted into this equation.
Φ_k	Shifted, floored softplus signal map with sharpness k.
T_{r}	Interval between opportunities to revise the command.
T_{obs}	Raw observation interval.
T_{assess}	Assessment interval.
τ_{dec}	Decision lag after assessment.
τ_{dep}	Deployment lag after formal decision.
τ_{eco}	Ecological response lag after implementation.
τ_m	Memory timescale of the filtered deficit signal.
$F^* = qE^*$	Model equilibrium fishing-mortality proxy in the illustrative core.
$\rho(T_{\mathrm{r}})$	Spectral radius of the local review-map Jacobian.

Symbol	Definition
A, B	Integrated periodogram power in 2.5–5-year and 5–9-year SSB bands.
A_{95}, B_{95}	Per-stock AR(1) 95% null thresholds for bands A and B.

S9 Screen sensitivity and stage-map reconstruction

S9.1 42-stock baseline and eleven-variant sensitivity battery

The baseline cohort contains 42 of 58 forage-fish series with at least 20 non-missing annual SSB observations. SSB is linearly detrended; normalized Lomb–Scargle power is integrated in bands A (2.5–5 years) and B (5–9 years). Each stock-specific AR(1) null uses 200 simulations and seed 7; the empirical tail probability is $(k + 1)/201$. Benjamini–Hochberg correction is applied to all 84 stock-by-band cells within each specification. In the baseline run, the minimum raw p-value is 0.0448 (PANCHNCHSP, band B) and the adjusted rejection count is zero.

Table S9: Eleven registered/recovered sensitivity specifications for the 42-stock screen. All have 84 tests and zero BH rejections; the minimum nominal p-value is reported to show resolution, not to select a winner.

Specification	Null / preprocessing	Minimum nominal p	BH rejections
Base	AR(1) surrogates; linear detrend	0.0448	0
ARMA	ARMA(1,1) surrogate; linear detrend	0.0697	0
Trend-stationary	Trend-stationary series; no detrending step	0.0299	0
Block 5	Circular block bootstrap, block length 5	0.0547	0
Block 10	Circular block bootstrap, block length 10	0.1692	0
Block 15	Circular block bootstrap, block length 15	0.2239	0
HP cycle	Hodrick–Prescott cycle plus AR(1) surrogate	0.0100	0
First difference	First-difference preprocessing plus AR(1)	0.0100	0
Regime median	Regime-median break surrogate	0.0547	0
Regime low	Regime-tercile-1 break surrogate	0.0498	0
Regime up	Regime-tercile-2 break surrogate	0.0746	0

The HP and first-difference variants have minimum attainable nominal p-values at 0.0100 at the retained simulation resolution; none of their cells survives BH correction. The result is a stable adjusted-zero decision under the specified battery, not a universal absence claim. The variant family includes distinct nulls and preprocessing, so it should not be interpreted as eleven independent replications.

An exploratory alternative-band check also returns zero BH-significant cells for a broad 2–10-year band and cadence bands 1–3, 3–8, and 8–20 years. An effort-proxy band of 8–80 years is unresolvable for every stock because no eligible series spans three times the upper period. These alternative-band outputs are screening checks, not preregistered tests of the institution-specific mechanism. The full per-stock SSB tables, q-values, null thresholds, and scripts are in the source bundle.

S9.2 Archived stage-response regions and separate τ_0 decomposition

The following stage-model response regions are transcribed from the earlier project record. Their original generating computation, solver configuration, and initial histories are unavailable; they are therefore archived and provisional, not re-executed v53 findings.

The same archive reports observable-specific dominant peaks near 4 years in anchovy biomass and 12 years in effort, and near 8 years in sprat biomass and 60 years in effort. Components of one stationary periodic orbit cannot have different fundamental periods; harmonics, modulation, transient spectral content, and measurement windows remain unresolved. Archived amplitudes are described only as percent-scale biomass and order-one effort excursions because the record does not say whether amplitude means half-range or peak-to-peak. The archive also reports robustness of the anchovy-class region to 30% multiplicative assessment error, but the generating record is not available for independent verification.

A separate stage-delay diagnostic treats g as maturation delay and τ as an institutional delay. It distinguishes raw characteristic-root crossings from crossings at parameter points classified as stable when $\tau = 0$. The table below is transcribed from `stage_tau0.log`; r-windows are finite-grid root-search summaries, and fish-range counts include only scanned $r \geq 0.2\,\mathrm{yr}^{-1}$. This is not the later preregistered two-stage Beverton–Holt reconstruction and is not a cross-operator causal comparison.

Table S10: Archived exploratory stage-map classifications. They are reported at their original status and are not validated by the new reconstruction.

Class	Archived trajectory classification	Status / scope
Anchovy	Persistent tail oscillation near $T_r = 3$–4 years; weak response at 2 years; annual-review trajectories converge across tested response values.	Solver and histories not attached.
Sprat	Persistent tail oscillation near $T_r = 6$–12 years.	Solver and histories not attached.
Cod	Convergence for every tested $T_r \in [1, 20]$ years.	Tested-grid record only; not a theorem for all histories.
Slow stock	For $r \in (0.01, 0.05)\,\mathrm{yr}^{-1}$, oscillation on part of the grid below about 20–30 years and convergence at longer intervals, with transition brackets about 30–50 years; archived cycle periods about 250–360 years.	Extremely long period; most governance records cannot resolve repeated cycles.

Table S11: Archived $\tau = 0$ decomposition by maturation delay g and controller gain η.

η	g (yr)	Raw r-window	$\tau = 0$-stable-only window	Raw fish-range counts	Institutional-only counts
0.914	0.5	[0.0079, 0.0225]	Empty	0	0
0.914	1	[0.0079, 1.5719]	[1.5719, 1.5719]	1	1
0.914	2	[0.0079, 0.8105]	[0.7865, 0.7865]	2	1
0.914	5	[0.0076, 0.3705]	[0.2660, 0.3285]	12	8
0.914	10	[0.0072, 0.7865]	[0.0873, 0.7865]	6	1
0.914	20	[0.0068, 0.7185]	Empty	9	0
3.0	0.5	[0.0068, 0.0646]	[0.0627, 0.0646]	0	0
3.0	1	[0.0068, 1.5719]	[0.0686, 1.5719]	1	1
3.0	2	[0.0068, 0.8608]	[0.0798, 0.8105]	7	5
3.0	5	[0.0066, 1.5719]	[1.5719, 1.5719]	29	1
3.0	10	[0.0064, 1.4361]	[0.3093, 1.4361]	23	9
3.0	20	[0.0062, 1.0628]	[0.3285, 0.4574]	32	12

The archived log's nonlinear RK4 check does not independently certify the claimed decomposition: its printed rightmost-root values are positive even for entries labelled as cases where $\tau = 0$ "should be stable." This internal inconsistency is unresolved; the decomposition table is retained as a provisional finite-grid diagnostic, not as a verified separation of biological cohort resonance from institutional delay. The full source log and the root-search script are included for audit.

S9.3 Pre-registered two-stage reconstruction

The original stage-map generating code, solver settings, and initial histories were not found in the repository. To avoid inventing a reconstruction as though it were the original object, a plan was frozen on 2026-09-01 before simulation. It specifies a new annual two-stage Beverton–Holt delayed-recruitment map with adults A, juveniles J, a recruitment lag, filtered signal Z, and held effort E. Ecological parameter values are in Table S7; the exact pre-registration, code, data tables, and logs are retained. No parameter was fitted to archived response windows.

The trajectory rule uses 2,000 review steps and the last 500 as the tail. Persistent oscillation is relative tail SD of adult abundance $\geq 2\%$, weak response is 0.1–2%, and convergence is below 0.1%. A separate finite integer grid records the local multiplier spectral radius. These two records are not interchangeable: a finite simulated trajectory classification and a local multiplier scan answer different questions, and the discrete stage map is not the three-state logistic continuous-flow map.

Table S12: Frozen comparison criteria and post-run verdicts for the new stage-map reconstruction. "Match" is agreement with the stated criterion on its declared grid; it does not validate the unavailable legacy solver.

Class	Legacy criterion	Reconstruction verdict
Anchovy	Persistent oscillation at $T_\mathrm{r} = 3$–4	Mismatch
Anchovy	Weak response at $T_\mathrm{r} = 2$, below the 3–4 window	Mismatch
Anchovy	Convergence at annual review	Match
Sprat	Persistent oscillation at $T_\mathrm{r} = 6$–12	Mismatch
Cod	Convergence for every integer $T_\mathrm{r} \in [1, 20]$	Match
Slow stock	Oscillation below roughly 20–30 years; convergence around 30–50 years	Match

The mismatch is reported without retuning. The archived window remains provisional, and a reconstruction mismatch does not adjudicate the legacy numbers because their generating object is unavailable. Full trajectory samples are `results/p5_stage_trajectories.csv`; equilibrium and multiplier records are in the adjacent `results/` files.

The local multiplier record at primary $q = 0.001$ has annual-review radii 0.8948 (anchovy), 0.9231 (sprat), 0.9560 (cod), and 0.9939 (slow stock). On the integer $T_\mathrm{r} = 1$–50 grid, the extractive multiplier crosses above unity between years 33 and 34 for anchovy and sprat, and between 34 and 35 for cod; the slow-stock multiplier remains below unity throughout. The protective channel remains stable on the integer grid. These linear records are distinct from the finite-horizon trajectory classifier: at $T_\mathrm{r} = 10$, the slow-stock trajectory is labelled persistent oscillation despite a stable local multiplier, leaving bistability, a non-decayed transient, or threshold sensitivity unresolved.

At the declared sensitivity $q = 0.1$, the extractive annual multipliers exceed one for all four classes ($\rho(1) = 2.229, 2.128, 1.904, 1.292$, respectively). On the integer grid, anchovy is below one at 19–35 years, sprat at 23–36, cod at 32–41, and slow stock remains above one; the protective channel is stable throughout. Crossings are bracketed by integer intervals, not refined to a continuous-time boundary. A declared steepness sensitivity $h = 0.6, 0.9$ is retained in `p5_stage_h_sensitivity.csv`; it is a local stability diagnostic, not the trajectory-classification criterion.

Under 30% multiplicative assessment error, anchovy and sprat remain converged at their tested legacy-window cells (maximum relative tail SD 0.00086). Cod is converged at $T_\mathrm{r} = 1, 5$ and weak at 10,20 years (maximum relative tail SD 0.00249). The slow-stock cells are converged at 5 years, weak at 10 and 20, and exceed the 2% persistence threshold at 30–50 years (0.02126–0.02645). Because the trajectory threshold is not noise-adjusted, these long-horizon labels may reflect noise-driven variance rather than a persistent deterministic oscillation. These checks remain specific to the reconstructed plant and do not validate the archived windows.

S10 Recovered T_r-ranked test: procedure, output, and limits

S10.1 Procedure and estimands

The executable `p7_empirical/tr_ranked_test.py` implements a pre-specified cross-sectional test on RAM v4.66 stocks. It uses the same series loader and spectral functions as the main stock screen: discard stocks with

fewer than 20 non-missing annual SSB observations; linearly detrend; compute a normalized Lomb–Scargle spectrum on the declared frequency grid; integrate powers in bands A (2.5–5 years), B (5–9 years), C (9–14 years), and D (14–30 years); fit a per-stock AR(1) null; and estimate 95% thresholds from 200 simulations with seed 7. The table retains A–D power but the ranked estimands below use bands A and B.

For each stock, the code assigns $T_r = 0.5$ years to the seven Peru/Chile annual-quota/in-season-closure group cases and $T_r = 1.0$ year to 35 annual-TAC proxy cases. The 4-year anchovy-class and 8-year sprat-class windows are both above either assigned value. The South African OMP's 4–5-year rule review is explicitly excluded from T_r; implementation lag is not T_r. The assignments are group-coded and not individually primary-verified.

The prespecified null-excess estimand is band power divided by that stock's AR(1) 95% threshold, A/A_{95} or B/B_{95}. Raw power is reported only as a confound diagnostic. Test 1 is Spearman association of the assigned T_r rank with each null-excess ratio; Test 2 is a two-sided Mann–Whitney comparison of the 0.5- and 1-year groups in band A, on raw power and null-excess; Test 3 counts nominal 95% threshold flags by broad stock class. The reported p-values are nominal and do not adjust for the several reported contrasts. Spearman has only two distinct T_r values and extensive ties; it is not a continuous dose-response test.

S10.2 Results and the corrected interpretation

The output contains 42 stocks: 7 at the subannual 0.5-year proxy and 35 at the annual 1-year proxy. Both groups are below the modeled 4- and 8-year windows. Therefore, this cohort has no observed proxy value near or beyond either window, and it cannot identify how the response changes as T_r approaches a window. The correct finding is **no identifiable window-approach gradient**.

The script does reproduce opposite-direction two-group associations. Median band-A null-excess is 0.529 in the 0.5-year group and 0.257 in the annual group (Mann–Whitney $U = 200$, $p = 0.007$). On raw band-A power the medians are 0.193 and 0.069 ($U = 219$, nominal $p < 0.001$). Spearman association is negative for band-A null-excess ($\rho = -0.408, p = 0.007$) and band-B null-excess ($\rho = -0.340, p = 0.028$); raw-power associations are $\rho = -0.509, p = 0.001$ for band A and $\rho = -0.393, p = 0.010$ for band B. These signs are opposite a simple prediction that excess rises as the review interval approaches the modeled response window: the shorter proxy group is farther below both windows and has the larger observed ratios.

These two-group associations are reproduced outputs, but they are not evidence of an institutional effect. All seven shorter-interval observations are Peru/Chile cases; proxy intervals are group-level and unverified stock by stock; other annual cases form the comparison group; regional ecology and data-generation differences are not randomized. ENSO-related quasi-periodicity is a plausible candidate confound, not a cause established by this test. AR(1) surrogates account for fitted red-noise persistence but do not remove quasi-periodic forcing. No causal adjustment, region-stratified estimate, or multi-year responsive comparison is available here.

At the nominal 95% threshold, there are 0/42 band-A flags and 1/42 band-B flag. By broad class, anchovy has 0/18 A and 1/18 B flags; sprat 0/4 in both; herring 0/20 in both. These are not BH-adjusted discoveries from a new test family. They agree with the main screen's adjusted-zero result, but are not independent replications.

S10.3 Per-stock values

Table S13 gives every stock's sample size, group-coded proxy, A/B band powers, per-stock AR(1) 95% thresholds, and nominal threshold flags. The null-excess ratios are calculated directly from the displayed powers and thresholds. The underlying CSV additionally retains regime and coding-source text.

Table S13: Per-stock output of the recovered and re-executed T_{r}-ranked test. Band power and AR(1) 95% threshold are shown separately; the null-excess ratio is their quotient. The T_{r} value is a group-coded proxy, not a stock-level verified field.

Stock ID	n	T_{r} (yr)	A	A_{95}	A flag	B	B_{95}	B flag
ANCHIXa	33	1.0	0.2663	0.4465	no	0.4306	0.4390	no
ANCHMEDGSA17-18	48	1.0	0.0091	0.1709	no	0.0541	0.2576	no
ANCHMEDGSA29	34	1.0	0.0264	0.2626	no	0.1097	0.3316	no
ANCHMEDGSA7	20	1.0	0.1933	0.5233	no	0.1977	0.5112	no
ANCHOBAYB	34	1.0	0.2408	0.4166	no	0.1400	0.4297	no
ANCHOSA	36	1.0	0.1263	0.3532	no	0.2441	0.3687	no
ARGANCHONARG	30	1.0	0.0930	0.3801	no	0.3203	0.4207	no
ARGANCHOSARG	30	1.0	0.1047	0.4073	no	0.3841	0.4347	no
HERR2224IIIa	31	1.0	0.0664	0.3267	no	0.1381	0.3919	no
HERR2529-32	48	1.0	0.0365	0.1945	no	0.0525	0.2563	no
HERR2532	38	1.0	0.0839	0.2337	no	0.0440	0.3069	no
HERR30	44	1.0	0.0601	0.3646	no	0.0682	0.3800	no
HERR30-31	61	1.0	0.0301	0.1486	no	0.0244	0.2130	no
HERR31	31	1.0	0.0755	0.3456	no	0.1581	0.3968	no
HERR4RFA	53	1.0	0.0251	0.1749	no	0.0231	0.1963	no
HERR4RSP	53	1.0	0.0409	0.1871	no	0.0939	0.2381	no
HERR4TFA	41	1.0	0.0463	0.2402	no	0.1088	0.3095	no
HERR4TSP	41	1.0	0.0317	0.2457	no	0.1237	0.3120	no
HERR4VWX	42	1.0	0.0319	0.2372	no	0.1596	0.3453	no
HERRIsum	37	1.0	0.0845	0.2966	no	0.1912	0.3437	no
HERRNIRS	63	1.0	0.0288	0.1538	no	0.0261	0.2222	no
HERRNORSS	74	1.0	0.0352	0.1142	no	0.0592	0.1520	no
HERRNS-IIIa-VIId	77	1.0	0.0272	0.1123	no	0.0399	0.1577	no
HERRNWATLC	58	1.0	0.0806	0.1907	no	0.0590	0.2376	no
HERRRIGA	47	1.0	0.0576	0.2838	no	0.1716	0.3372	no
HERRSIRS	64	1.0	0.0615	0.1751	no	0.0466	0.2172	no
HERRVIa	57	1.0	0.0594	0.1884	no	0.0554	0.2473	no
HERRVIaVIIbc	62	1.0	0.0694	0.2127	no	0.0819	0.2685	no
JANCHOPJPN	36	1.0	0.1744	0.3416	no	0.1181	0.3666	no
JANCHOSETO	38	1.0	0.2361	0.5016	no	0.1117	0.4083	no
JANCHOTSST	42	1.0	0.0872	0.2987	no	0.1959	0.3861	no
PANCHCCH	36	0.5	0.3217	0.4890	no	0.2913	0.4082	no
PANCHCHVX	23	0.5	0.1123	0.4350	no	0.1433	0.4433	no
PANCHCSCH	31	0.5	0.1046	0.3494	no	0.1116	0.3997	no
PANCHNCH	29	0.5	0.4633	0.6086	no	0.2410	0.3586	no
PANCHNCHSP	36	0.5	0.1927	0.4796	no	0.4247	0.4075	yes
PANCHPERUNC	47	0.5	0.1471	0.2782	no	0.1726	0.3290	no
PANCHSCH	28	0.5	0.2715	0.4673	no	0.3221	0.4459	no
SPRAT22-32	50	1.0	0.0978	0.2209	no	0.0464	0.2823	no
SPRATIIIa-IV	48	1.0	0.1030	0.2150	no	0.1004	0.2731	no
SPRATNS	45	1.0	0.2171	0.3542	no	0.0954	0.3690	no
SPRBLKGSA29	30	1.0	0.1014	0.3993	no	0.3536	0.4312	no

S10.4 Reproduction and status reconciliation

The recovered output table and reproduction log match the script to the reported precision. The historical query-log entry R4 records that, at that time, a search found only a recommendation memo marked "never executed." Later repository state contains the implemented script and archived output. The v53 reproduction was run against the current local RAM v4.66 series and produced the same 42-stock output and test summary. The stale memo is retained in the provenance record; it does not overwrite the executed script/log evidence. The source bundle carries the script, CSV, stdout, and a reconciliation note.

S11 Logistic core: crossing, precision, and parameter sensitivity

S11.1 Equilibrium and linear map

The logistic core uses $S(N) = rN(1 - N/K)$ and the zero-deficit equilibrium $Z^* = \delta = \log(2)/10$. Solving the interior root of the effort law yields $E^* = 2.089623397737668$; the stock equilibrium is $N^* = K(1 - qE^*/r) = 89.55188301131166$. The equilibrium fishing mortality is $F^* = qE^* = 0.002089623397737668\,\mathrm{yr}^{-1}$. Local effort gains are $(C_E, C_Z) = (-0.0595180098426, 1.7850187149850)$ for the mobilising law and $(-0.850336140482, -1.661701896635)$ for the protective comparator.

For held effort, the plant's linearisation is the matrix A_{hold} retained in `work/results.json` and regenerated by `work/p07_core.py`. The review matrix composes the held plant flow with either (i) the exponential frozen-assessment controller-flow update, whose local linearization is exact for the controller ODE with the endpoint assessment fixed, or (ii) the separately defined forward-Euler update. Eigenvalues are calculated for the product in the registered flow-then-update ordering. The full state dimension is three; no continuous-delay approximation is used in this matrix calculation.

S11.2 Baseline operator record

Table S14: Baseline radii at annual review and all computed crossings on the interval scan. Values are from the retained full-precision core output.

Map	$\rho(1\,\mathrm{yr})$	Crossing record / grid interpretation
Mobilising, exponential frozen assessment	1.0003506565	One complex-pair crossing at 6.5012949481 yr, unstable → stable.
Mobilising, Euler command step	1.0005452356	Complex-pair crossing at 47.5359822758 yr, unstable → stable; real -1 crossing at 79.1426878068 yr, stable → unstable.
Protective, exponential frozen assessment	0.9838095183	No crossing on $[0.2, 200]$ yr; maximum sampled $\rho = 0.996740$.
Protective, Euler command step	0.9837962340	Real -1 crossing at 2.3063544244 yr, stable → unstable.

At the exponential-update mobilising crossing, the unit-modulus pair is approximately $0.9846404664 \pm 0.1745942495i$, with angle 0.175494 radians. The exponential-update crossing estimate is unchanged to the printed digits on 2,001-, 20,001-, and 200,001-point preliminary scans; each reported value is then refined by 70 bisection iterations. The sign of $\rho(1) - 1$ is only about 3.5066×10^{-4} at baseline. Euler crossings at 47.536 and 79.143 years, and the Euler protective -1 crossing at 2.306 years, are properties of the Euler update; they are not numerical approximations to the exponential-update boundary at those intervals.

A continuous-delay comparator from the archived companion calculation has approximate crossings at 3.666 and 150.358 years. It is retained as an operator contrast, not as a third discretization of the same review map. The exponential frozen-assessment result and the continuous-delay result have different state/operator constructions; their separation cannot be attributed to a single parameter while changing all architecture at once.

S11.3 Re-run one-at-a-time $\pm 2\%$ sensitivity

The most sensitive directions among this scalar battery are $K, q, E_{\max}, \Delta_{\mathrm{ref}}$. A missing crossing is a qualitative change in this scanned result, not simply a displaced point estimate. The signs are not symmetric: for example, a 2% increase in Δ_{ref} removes the crossing from the scan, while a 2% decrease moves it to 10.775 years. The table is deterministic parameter perturbation, not statistical uncertainty.

Table S15: Re-run first exponential-update mobilising crossing (years) after one-at-a-time changes in each parameter. "None" means no crossing was found on the registered $[0.2, 200]$-year scan.

Parameter	-2%	-1%	$+1\%$	$+2\%$
K	None	2.986	8.665	10.371
q	None	3.438	8.502	10.100
$E_{\max}$	None	3.445	8.499	10.096
η	5.270	5.924	7.021	7.494
r	7.349	6.936	6.041	5.549
Δ_{ref}	10.775	8.891	2.353	None
τ_m	6.599	6.551	6.449	6.394
δ_0	6.495	6.498	6.505	6.508
Z_{ref}	6.506	6.504	6.499	6.497

S11.4 Archived one-at-a-time $\pm 5\%$ battery

A separate archived sensitivity battery varied ten parameters at factors 0.90, 0.95, 1.05, and 1.10. It is retained as an archived record, not described as the current v53 rerun. The crossing-only summary is reproduced from `src/p7_empirical/recovered/sens_table.csv`; the file also retains equilibrium values, annual radii, crossing angles, crossing types, and protective-grid maxima. "None" denotes no crossing on the archived interval scan. The re-run $\pm 2\%$ table above and this archived wider perturbation record are distinct analyses.

Table S16: Archived one-at-a-time $\pm 5\%$ and $\pm 10\%$ sensitivity battery. Entries are first exponential-update mobilising crossings in years; "None" means no crossing on the archived interval scan.

Parameter	-10%	-5%	$+5\%$	$+10\%$
r	10.1844	8.4928	3.7490	None
K	None	None	14.2825	19.0977
q	None	None	13.7973	18.3769
$E_{\max}$	None	None	13.7899	18.3665
η	None	2.1944	8.7154	10.3140
δ_0	6.4676	6.4845	6.5181	6.5348
Z_{ref}	6.5268	6.5136	6.4897	6.4788
Δ_{ref}	20.8565	15.1725	None	None
τ_m	6.9012	6.7279	6.2138	5.8545
δ	None	4.6883	7.6905	8.5444

S11.5 Six-parameter joint perturbation: 20,001-point rerun

The six jointly perturbed parameters are $(r, K, q, \eta, E_{\max}, \Delta_{\mathrm{ref}})$, each independently set to the lower or upper edge of its stated relative box, producing 64 corners per box. Both boxes were recomputed from scratch on 20,001 evenly spaced review intervals in $[0.2, 200]$ years (grid spacing about 0.00999 years) and crossings were bisected. The full per-corner JSON is retained as `work/joint_box_fullgrid.json`.

Table S17: Full-grid joint-box results. "Stable no crossing" means $\rho < 1$ at all 20,001 points; it is not a proof of stability on the continuum.

Box	Corners	Crossing	No crossing	Crossing range (yr)	Max ρ among no-crossing corners
$\pm 0.5\%$	64	60	4	0.871965–10.670425	0.9999712801
$\pm 1\%$	64	44	20	4.423192–13.653242	0.9999952956

Every no-crossing corner in both boxes is stable at all evaluated grid points; no no-crossing corner is grid-unstable. Because the maximum radii are close to one, the status is numerically delicate. These boxes are not posterior or confidence intervals, and no sampling distribution is assigned to the parameters. The results show that a single baseline crossing should not be reported as a robust institutional threshold.

S11.6 Independent numerical and nonlinear checks

The retained precision-certificate log reports a high-precision root $T_{\mathrm{r},c} = 6.501294948075468368899748616 84$, consistent with the double-precision monodromy crossing. A fresh independent audit of the full nonlinear plant and nonlinear frozen-assessment controller flow is in `work/nonlinear_flow_check_v53.py` and its JSON output. With DOP853 integration and central differences, the full-map Jacobian agrees with the monodromy to

maximum absolute error 1.67×10^{-9} at the crossing for relative step 3×10^{-5}; finite-difference crossing estimates across five steps range from 6.501276 to 6.501299 years. The prior normal-form script used a linearized controller update and therefore does not establish a normal-form coefficient for the full nonlinear map; its supercritical label is not relied on here. No invariant-circle continuation or independent orbit confirmation was completed, so no nonlinear bifurcation theorem is claimed.

S11.7 F-grid placement convention

The model's equilibrium fishing-mortality proxy is swept as $F^* = qE^*$ within its illustrative parameter grid. For the cross-system placement summaries in S13.2, the code uses fixed categories: below-stable if $F < 0.001$; marginal if $0.001 \leq F \leq 0.004$; unstable-range if $0.004 < F \leq 0.02$; beyond-feasible if $F > 0.02$. These names describe placement relative to the model grid and its maximum intrinsic growth rate, not assessed stock status. Their ecological scale is not calibrated to the 42-stock cohort; treat the categories as exploratory diagnostics only.

S12 Archived nonlinear comparator and trajectory-distortion records

S12.1 Executed logistic-core comparator design

The retained comparator script simulates the same illustrative logistic plant with the four feedback/update combinations (mobilising or protective; forward Euler or nonlinear frozen-assessment controller flow) and a fixed-plan controller with effort frozen at E^*. The initial state is $N_0 = 0.95N^*$, $Z_0 = Z^* = \delta$, and $E_0 = E^*$; the horizon is 800 reviews; metrics use the last 60% of simulated reviews; and divergence is recorded if the stock leaves $(0, 2K)$. The stock follows the held logistic flow; the memory signal is integrated with RK4 at $dt = 0.2$. The noisy run perturbs the assessment as $\widehat{Z} = Z(1 + \epsilon\xi)$, with standard-normal ξ, as implemented in the retained script. The intended robustness value is $\epsilon = 0.3$.

A single-seed deterministic/noisy table (`p5_comparator_mse.csv`) and a ten-seed summary (`p5_comparator_mse_mc.csv`) are archived. The ten-seed CSV has no ϵ, per-seed, or seed-map columns; the earlier manuscript labels it as the 30% error run. We therefore retain the 10-seed figures as archived, not as a v53 rerun, and flag the missing table metadata. Metrics are normalized by K for stock deviation and by $E_{\max}$ for effort deviation; depletion frequency is the fraction of post-transient reviews with $N < 0.5N^*$.

Table S18: Archived 10-seed nonlinear comparator summary. The retained CSV does not contain an ϵ or per-seed field; the earlier manuscript describes this summary as the 30% assessment-error run. Values are preserved as archived and are not presented as a new re-execution.

Controller	T_r	Seeds	Mean depletion	Mean min N/K	Mean RMSD N/K	Diverged seeds
extractive-euler	1.0	10	0.2179	0.2229	0.3156	0
extractive-exact	1.0	10	0.066	0.5406	0.1402	0
protective-euler	1.0	10	0.0	0.895	0.0002	0
protective-exact	1.0	10	0.0	0.895	0.0002	0
fixed	1.0	10	0.0	0.8954	0.0	0
extractive-euler	8.0	10	–	–	–	10
extractive-exact	8.0	10	0.0823	0.2608	0.1996	0
protective-euler	8.0	10	0.0	0.5952	0.275	0
protective-exact	8.0	10	0.0	0.8939	0.0005	0
fixed	8.0	10	0.0	0.8955	0.0	0
extractive-euler	12.0	10	–	–	–	10
extractive-exact	12.0	10	0.0315	0.2997	0.1524	0
protective-euler	12.0	10	0.9996	0.3769	0.4854	0
protective-exact	12.0	10	0.0	0.8935	0.0006	0
fixed	12.0	10	0.0	0.8955	0.0	0
extractive-euler	20.0	10	–	–	–	10
extractive-exact	20.0	10	0.001	0.5123	0.0662	0
protective-euler	20.0	10	–	–	–	10
protective-exact	20.0	10	0.0	0.8928	0.0008	0
fixed	20.0	10	0.0	0.8955	0.0	0

The archived ten-seed summary reports divergence in all 10 Euler-mobilising runs at $T_r = 8, 12, 20$ years; the Euler-protective controller diverges in all 10 runs at 20 years and has mean depletion 0.9996 at 12 years. The nonlinear frozen-assessment-flow protective and fixed-plan runs have zero recorded depletion at the displayed intervals. Nonlinear frozen-assessment-flow mobilising is intermediate: it avoids divergence in these cells, but the mean depletion frequencies are nonzero (for example 0.066 at one-year review and 0.0823 at eight years). This is a model-level robustness comparison, not evidence that one real policy class dominates across stocks or operating models.

S12.2 Linear trajectory-distortion record

The retained `p5_linear_trajectory_mse.csv` compares Euler and exponential-update linear trajectories under the same initial perturbation and interval grid. In the mobilising channel, the scaled RMSD is 1.913 at $T_r = 0.5$ years, 13.111 at 1 year, 189.020 at 2 years, and exceeds 10^4 for many longer intervals; Euler trajectories diverge in several intervals where the exponential-update map is stable. In the protective channel, the two updates are close at 0.5–2 years (scaled RMSD 0.0028–0.0172), but Euler divergence occurs from 3 years upward in the retained scan while the nonlinear frozen-assessment-flow trajectory remains bounded. The norm is defined by the archived analysis and should not be interpreted as a universal operational loss function.

Both trajectory records explain why the linear crossing comparison matters, but their status is narrower than the main spectral computation: one uses an illustrative plant and fixed perturbation; the noisy multi-seed table has a missing metadata field; and neither tests field performance. These results are not included in the main-text abstract or used to recommend a policy.

S13 ADH, fishing-mortality placement, and northern-cod records

S13.1 ADH buffer-years recomputation

The archived ADH measure is defined stock by stock as

$$\mathrm{ADH} = \max\left\{0, \frac{\ln(\mathrm{SSB}/B_{\mathrm{lim}})}{F}\right\},$$

where SSB is the current assessed spawning-stock biomass, B_{lim} is the stock-specific limit reference point, and F is the current fishing mortality. It is an algebraic buffer-years proxy, not a direct measurement of institutional response or a guarantee of time to recovery.

The recovered input cohorts include an archived 43-stock table, a 454-stock RAM v4.66 broader pool, a 415-row v4.44 pool, and the screen-cohort summaries. Recalculation matches deposited ADH to numerical precision (maximum absolute difference 1.14×10^{-13} in the v4.66 broad pool; the archived 43-stock values match exactly to printed precision). Descriptive medians and zero fractions reproduce: 43-stock archived median 1.7902 years with 18.60% zero values; v4.66 broad-pool median 3.3893 years with 15.20% zero values; v4.66 forage-fish subset median 2.6962 years with 18.75% zero values. In the 33-stock paired vintage comparison, the v4.44 and v4.66 medians are 2.7901 and 2.3442 years, respectively; across 396 paired broader stocks, median absolute ADH change is 0.7958 years. Data vintage and cohort definition therefore matter.

A fresh 20,000-draw seed-11 resampling run from the v4.66 broad pool yields median-of-pool draws with mean 3.3452 years and percentile range 1.8423–5.5307 years; 2.33% of draws fall below the archived 43-stock median of 1.7902. The archived resampling record gives 2.12%. The forage subset's new seed-11 run has median-of-pool draw mean 2.7916 years, percentile range 2.3081–3.5649, and 0.39% below the archived median (archived value 0.29%). This resampling describes a particular cohort-vintage construction; it is not a probability that a given stock is sustainable.

S13.2 Fishing-mortality data verification and placement

The local F script first checks three gates. (G1) It reproduces the screen cohort: 42 stocks with at least 20 valid SSB observations. (G2) It compares fisheries-ADH F with both mean F and last F: the archived fisheries-ADH values match last F for 42 of 43 intersected stocks; `SPRBLKGSA29` is the exception (archived fisheries-ADH F 0.2100, series mean F 0.2344, and last-F 0.0400, with 32 F observations). (G3) It reproduces the v4.66 broader-pool F quartiles (0.0800, 0.2143, 0.4639) and all 13 taxonomic-group quartile triples. These are arithmetic and source-cohort checks.

Among 38 of the 42 screen stocks with usable F series, the mean of stock-level F medians is 0.3663 (IQR 0.2701–0.7243; range 0.0053–2.6776). In the 454-stock broader pool, F_{now} has median 0.2143 (IQR 0.0800–0.4639). The 1170 paired TC/SSB and F observations have Spearman $\rho = 0.8243$, $p = 7.88 \times 10^{-291}$, showing that the two exploitation summaries are strongly associated in this merged record. A comparison of model

$F^* = 0.00209$ with empirical F values places most empirical values outside the illustrative model's low-F grid; because the model scale is not calibrated to these stocks, this is an explicit scale-mismatch warning rather than evidence that real fisheries are infeasible.

The exploratory rank association between stock-level mean F and target-band null-excess is positive but not conventionally significant in the screen cohort: band A $\rho = 0.3069, p = 0.0609$, band B $\rho = 0.2017, p = 0.2247$. Excluding the seven Peru/Chile cases reduces these to band A $\rho = 0.0339, p = 0.8565$ and band B $\rho = 0.0327, p = 0.8615$. The attenuation is consistent with region-linked confounding, but it does not identify the confound or establish a causal effect. The per-stock F series and model-placement labels are shown below.

Table S19: Screen-cohort fishing-mortality series and placement categories. Values are retained as the local v53 output; a dash indicates no usable F series.

Stock ID	SSB n	F n	Mean F	Median F	Max F	Placement
ANCHIXa	33	32	1.0000	0.9697	1.8305	beyond-feasible
ANCHMEDGSA17-18	48	48	0.6385	0.5540	1.5400	beyond-feasible
ANCHMEDGSA29	34	34	1.2613	1.1854	1.9805	beyond-feasible
ANCHMEDGSA7	20	20	0.1085	0.1060	0.2310	beyond-feasible
ANCHOBAYB	34	27	0.3336	0.3640	0.6810	beyond-feasible
ANCHOSA	36	0	-	-	-	no-F
ARGANCHONARG	30	30	0.0437	0.0420	0.0960	beyond-feasible
ARGANCHOSARG	30	30	0.0053	0.0040	0.0140	unstable-range
HERR2224IIIa	31	29	0.4220	0.4070	0.7010	beyond-feasible
HERR2529-32	48	47	0.2451	0.2190	0.4600	beyond-feasible
HERR2532	38	37	0.2701	0.2430	0.4820	beyond-feasible
HERR30	44	42	0.1411	0.1425	0.2060	beyond-feasible
HERR30-31	61	60	0.0841	0.0587	0.2564	beyond-feasible
HERR31	31	31	0.3294	0.2880	0.6400	beyond-feasible
HERR4RFA	53	30	0.0841	0.0705	0.1990	beyond-feasible
HERR4RSP	53	39	0.2128	0.1750	0.5780	beyond-feasible
HERR4TFA	41	36	0.3270	0.3275	0.9060	beyond-feasible
HERR4TSP	41	36	0.4930	0.4240	1.7400	beyond-feasible
HERR4VWX	42	41	0.7083	0.5700	2.4600	beyond-feasible
HERRIsum	37	31	0.2701	0.2600	0.4400	beyond-feasible
HERRNIRS	63	62	0.4352	0.3830	0.9930	beyond-feasible
HERRNORSS	74	73	0.2945	0.1500	3.4900	beyond-feasible
HERRNS-IIIa-VIId	77	76	0.3384	0.2520	1.2100	beyond-feasible
HERRNWATLC	58	57	0.3777	0.2750	0.7980	beyond-feasible
HERRRIGA	47	45	0.4185	0.4240	0.7080	beyond-feasible
HERRSIRS	64	63	0.4734	0.4500	0.9740	beyond-feasible
HERRVIa	57	57	0.3549	0.2720	1.0800	beyond-feasible
HERRVIaVIIbc	62	61	0.3409	0.2760	1.0400	beyond-feasible
JANCHOPJPN	36	0	-	-	-	no-F
JANCHOSETO	38	0	-	-	-	no-F
JANCHOTSST	42	0	-	-	-	no-F
PANCHCCH	36	36	0.5616	0.5025	1.7000	beyond-feasible
PANCHCHVX	23	23	1.3357	1.3700	3.1100	beyond-feasible
PANCHCSCH	31	31	1.2472	1.0500	3.6200	beyond-feasible
PANCHNCH	29	15	2.6776	2.9800	4.0200	beyond-feasible
PANCHNCHSP	36	36	2.6179	1.9250	7.9800	beyond-feasible
PANCHPERUNC	47	47	0.7297	0.6150	2.1000	beyond-feasible
PANCHSCH	28	28	0.8514	0.7800	2.7800	beyond-feasible
SPRAT22-32	50	48	0.3179	0.3189	0.5320	beyond-feasible
SPRATIIIa-IV	48	47	1.2363	1.2750	2.2030	beyond-feasible
SPRATNS	45	44	1.0598	1.0185	2.3200	beyond-feasible
SPRBLKGSA29	30	32	0.2344	0.2050	0.6400	beyond-feasible

S13.3 Northern-cod assessment table

Table S20: Northern-cod assessment values shown in the descriptive two-window figure. The reported instantaneous natural mortality M comes from the assessment table; e^{-M} is a transformation, not a separate observed survival series.

Year	SSB (kt)	M (yr^{-1})	e^{-M}
1991	735.00	1.002	0.367
1992	382.00	2.214	0.109
1993	101.00	2.575	0.076
1994	31.00	2.331	0.097
1995	10.00	0.288	0.750
1996	16.05	0.341	0.711
2000	34.42	0.717	0.488
2004	20.07	0.362	0.696
2005	25.18	0.288	0.750
2010	96.91	0.696	0.499
2015	298.65	0.278	0.757

The values were cross-checked against the DFO CSAS SAR 2016/026 table. The crash window is 1991–1995; the later plotted interval extends to 2015 and includes low-stock persistence and subsequent increases. The plot partitions the chronology but does not decompose causation. No numerical constrained-M allocation is included in v53: the raw source quantities and a fully reproducible calculation were not recovered. Claims about the size of a seal-attributable, fishing-attributable, or governance-attributable component are therefore not made.

S14 Power record and unreproduced items

S14.1 Injected-signal power: 17 archived white-noise cells

The committed driver specifies a sampled-governance operating model, generates annual effort, adds multiplicative observation error $E_t^{\text{obs}} = E_t \exp(\varepsilon_t)$, and tests target-band periodogram power against a per-series AR(1) red-noise null. The null uses 120 simulations and seed 7. Each power cell has 50 trial seeds 0–49. White-noise draws use $\varepsilon_t \sim N(0, \sigma^2)$, with $\sigma \in \{0.1, 0.3\}$. The analyzed record spans H years after burn-in; the cell-specific integrator horizon is retained in the CSV.

The model cells are: sprat class $(r, g, T_r) = (0.8, 2, 7)$ with target band 30–120 years; anchovy class $(1.6, 1, 3)$ with band 8–20 years; and a cod-labelled false-positive class $(0.3, 5, 5)$ with band 8–20 years. The archived 17-cell design is not a complete Cartesian grid: sprat $H = 400, \sigma = 0.1$ is not among the specified cells. Fifteen cell outputs are present in the recovered run log; the two cod $H = 400$ values were recovered from a follow-up JSON. The complete driver previously stopped after 15/17 cells, so the consolidated table is an assembly of archived outputs, not a claim of a successful full driver rerun.

Table S21: Archived white-noise injected-signal power table (17 cells). Each cell has 50 trials; power is detections divided by trials.

Class	H	Horizon T	σ	Detections / 50	Power
sprat	100	120	0.1	50 / 50	1.00
sprat	100	120	0.3	12 / 50	0.24
sprat	200	220	0.1	50 / 50	1.00
sprat	200	220	0.3	35 / 50	0.70
sprat	400	420	0.3	50 / 50	1.00
anchovy	100	120	0.1	5 / 50	0.10
anchovy	100	120	0.3	4 / 50	0.08
anchovy	200	220	0.1	5 / 50	0.10
anchovy	200	220	0.3	3 / 50	0.06
anchovy	400	420	0.1	0 / 50	0.00
anchovy	400	420	0.3	4 / 50	0.08
cod	100	120	0.1	4 / 50	0.08
cod	100	120	0.3	4 / 50	0.08
cod	200	220	0.1	2 / 50	0.04
cod	200	220	0.3	4 / 50	0.08

Class	H	Horizon T	σ	Detections / 50	Power
cod	400	420	0.1	1 / 50	0.02
cod	400	420	0.3	2 / 50	0.04

The sprat signal is detected in 50/50 trials at $H = 100$ and 200 when $\sigma = 0.1$; at $\sigma = 0.3$, power rises from 12/50 to 35/50 across those horizons, and is 50/50 in the single retained $H = 400$ cell. Anchovy power ranges 0/50–5/50 (0.00–0.10), and the cod-labelled false-positive rate ranges 1/50–4/50 (0.02–0.08). These conditional proportions are based on 50 trials and the declared model/null pairing. They do not certify adequate power for empirical fisheries or a specific stock. The sprat-class high detection at low noise can be driven by a growth transient in a broad 30–120-year band, not necessarily a stable cycle; power falls substantially at higher noise and shorter records. The empirical screen's null therefore cannot be interpreted as a strong mechanism test for every ecological class.

S14.2 Northern-cod constrained-mortality formulation: not reproduced

The constrained-M formulation in earlier project notes proposes a mortality allocation that could distinguish fishing from other losses. In v53, its underlying table, uncertainty structure, and complete executable calculation were not recovered. The DFO assessment table used here reports SSB and natural mortality estimates, but does not by itself identify the causes of mortality. Consequently, v53 does not report numerical constrained-M estimates, attributed mortality shares, or a causal seal-versus-fishing decomposition. Those values must not be reconstructed from the plotted series alone. The open item is to recover the original input table, document the model definition and priors/constraints, and reproduce the calculation against alternative assessment formulations before drawing a mortality-allocation conclusion.

S14.3 Prospective work is not completed work

The five research designs and their estimands, identification checks, and falsification rules are specified in S7. The source bundle also retains model-comparison, governance-event, human-in-the-loop, and MSE proposals. None has an executed data analysis, registered protocol identifier, field intervention, or effect estimate in v53. The 17-cell conditional power table can inform a future simulation plan; it is not itself a completed prospective field test.